

Unit – III:

Regressions with Qualitative Independent Variables

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Unit Objective

- Understand and gain the knowledge regarding advanced regression models and its application for decision making.

Unit Contents

- **Regressions with Qualitative Independent Variables:**
 - Dummy variable technique – Use of dummy variables, interaction effects, Testing structural stability of regression models,
 - Comparing two regressions, seasonal analysis, piecewise linear regression Regression with dummy dependent variables. The LPM, Logit, Probit and Tobit models – Applications.

Qualitative / Dummy Variables

- Qualitative variables are nominal scale variables which have no particular numerical values.
- We can “quantify” them by creating the so-called dummy variables, which take values of 0 and 1
 - 0 indicates the absence of an attribute
 - 1 indicates the presence of the attribute
- For example, a variable denoting gender can be quantified as female = 1 and male = 0 or vice versa.
- Dummy variables are also called indicator variables, categorical variables, and qualitative variables.
 - *Examples:* gender, race, color, religion, nationality, geographical region, party affiliation, and political upheavals

Dummy Variable Trap

- If an intercept is included in the model and if a qualitative variable has m categories, then introduce only $(m - 1)$ dummy variables.
- For example, suppose we are interested in political affiliation, a categorical variable that might assume three values - Republican, Democrat, or Independent. We could represent political affiliation with two dummy variables:
 $X_1 = 1$, if Republican; $X_1 = 0$, otherwise.
 $X_2 = 1$, if Democrat; $X_2 = 0$, otherwise.
- In this example, notice that we don't have to create a dummy variable to represent the "Independent" category of political affiliation. If X_1 equals zero and X_2 equals zero, we know the voter is neither Republican nor Democrat. Therefore, voter must be Independent.
- If we do not follow this rule, we will fall into what is called the dummy variable trap, the situation of perfect collinearity.

Problem 1

- Consider the table below. It uses three variables to describe 10 students. Two of the variables (Test score and IQ) are quantitative. One of the variables (Gender) is categorical.

Student	Test score	IQ	Gender
1	93	125	Male
2	86	120	Female
3	96	115	Male
4	81	110	Female
5	92	105	Male
6	75	100	Female
7	84	95	Male
8	77	90	Female
9	73	85	Male
10	74	80	Female

For this problem, we want to test the usefulness of IQ and Gender as predictors of Test Score. To accomplish this objective, we will:

Problem 1

- Recode the categorical variable (Gender) to be a quantitative, dummy variable.
- Define a regression equation to express the relationship between Test Score, IQ, and Gender.
- Conduct a standard regression analysis and interpret the results.
- Therefore, we can express the categorical variable Gender as a single dummy variable (X_1), like so:
 - $X_1 = 1$ for male students.
 - $X_1 = 0$ for non-male students.
- Now, we can replace Gender with D_1 in our data table.

Problem 1

Student	Test score	IQ	D1
1	93	125	1
2	86	120	0
3	96	115	1
4	81	110	0
5	92	105	1
6	75	100	0
7	84	95	1
8	77	90	0
9	73	85	1
10	74	80	0

For this problem, the equation is:

$$\hat{Y} = b_0 + b_1 IQ + b_2 D_1$$

where $\hat{Y}$ is the predicted value of the Test Score, IQ is the IQ score, D_1 is the dummy variable representing Gender, and b_0 , b_1 , and b_2 are regression coefficients.

Problem 1

Data Analysis

To complete a good multiple regression analysis, we want to do four things:

- Estimate regression coefficients for our regression equation.
- Assess how well the regression equation predicts test score, the dependent variable.
- Assess the extent of multicollinearity between independent variables.
- Assess the contribution of each independent variable (i.e., IQ and Gender) to the prediction.

Problem 1

Regression Coefficients

	<i>Coef</i>	<i>Std err</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	38.6	9.535	4.048	0.005
IQ	0.4	0.094	4.278	0.004
Gender	7	2.686	2.606	0.035

$$\hat{y} = 38.6 + 0.4 * IQ + 7 * D_1$$

The multicollinearity problem is not present.

Here, the regression coefficient for gender is 7. This suggests that, after effects of IQ are taken into account, males will score 7 points higher on the test than the reference group (females). And, because the regression coefficient for gender is statistically significant, we interpret this difference as a real effect - not a chance artifact.

SUMMARY OUTPUT	
<i>Regression Statistics</i>	
Multiple R	0.900113
R Square	0.810203
Adjusted R Square	0.755975
Standard Error	4.181592
Observations	10

Interaction Effects in Regression

- In regression, an interaction effect exists when the effect of an independent variable on a dependent variable changes, depending on the value(s) of one or more other independent variables.

Interaction Effects in Equations

- In a regression equation, an interaction effect is represented as the product of two or more independent variables. For example, here is a typical regression equation *without* an interaction:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2$$

where $\hat{Y}$ is the predicted value of a dependent variable, X_1 and X_2 are independent variables, and b_0 , b_1 , and b_2 are regression coefficients.

- And here is the same regression equation *with* an interaction:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_1X_2$$

- Here, b_3 is a regression coefficient, and X_1X_2 is the interaction. The interaction between X_1 and X_2 is called a two-way interaction, because it is the interaction between two independent variables.

Regression Analysis With Interaction Terms

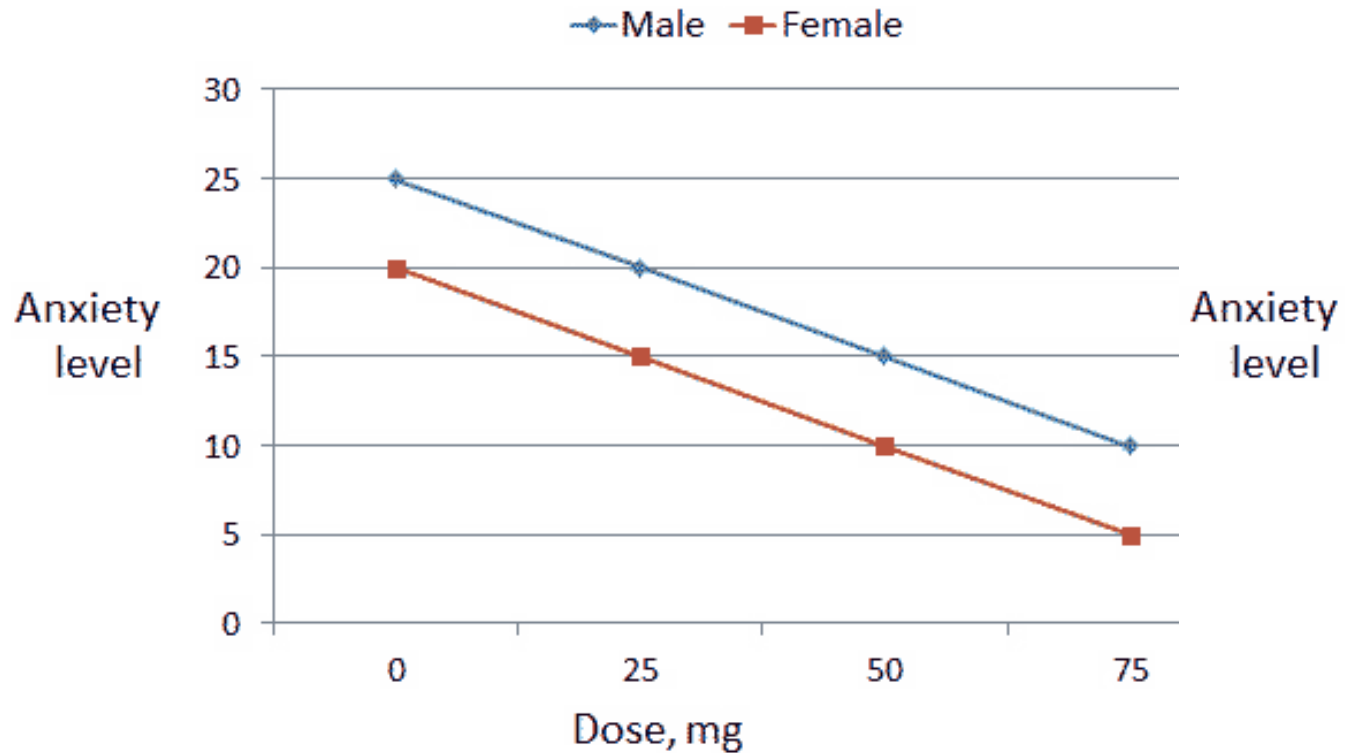
- Consider the problem where the goal is to examine effects of drug dosage and gender on anxiety levels. The data is presented as below: [Male=1, 0 otherwise]

Anxiety	Dose, mg	Gender	D*G
90	0	0	0
88	20	0	0
85	40	0	0
83	60	0	0
88	0	1	0
72	20	1	20
62	40	1	40
43	60	1	60

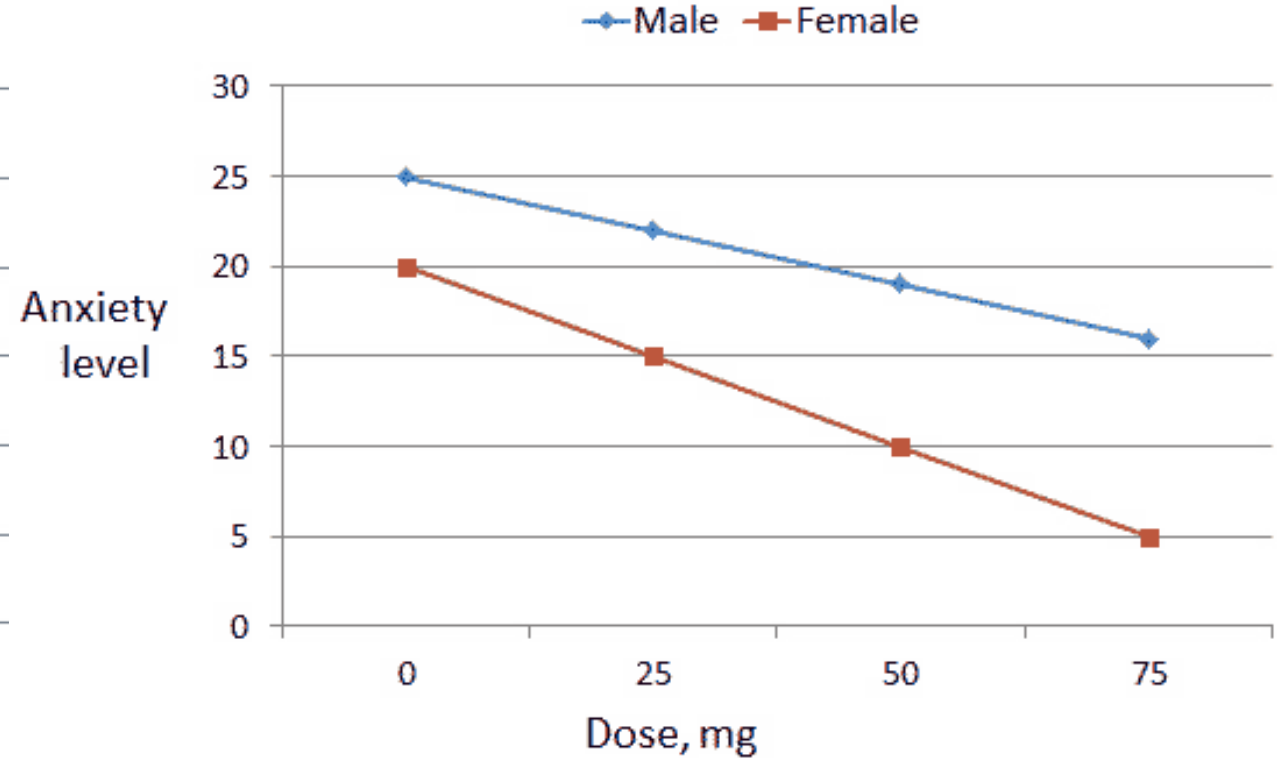
To understand potential interaction effects, compare the lines from the interaction plot:

- If the lines are parallel, there is no interaction.
- If the lines are not parallel, there is an interaction.

Regression Analysis With Interaction Terms



In the plot above, the lines are parallel. This suggests no interaction effect, based on gender.



In the plot above, the lines are not parallel. This suggests possible interaction effect, based on gender.

Regression Analysis With Interaction Terms

- Without Interaction

	<i>Coef</i>	<i>Std Err</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	99.175	6.006	16.512	0.000
Dose	-0.423	0.138	-3.066	0.028
Gender	-20.250	6.162	-3.286	0.022

Coefficient of multiple determination is a respectable 0.80.

- With Interaction

	<i>Coef</i>	<i>Std Err</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	90.100	1.548	58.190	0.000
Dose	-0.120	0.041	-2.900	0.044
Gender	-2.100	2.190	-0.959	0.392
DG	-0.605	0.059	-10.338	0.000

Coefficient of multiple determination is 0.99.

We see that the interaction between dose and gender is statistically significant at the 0.01 level. When we examine the main effects, we see that dose is statistically significant, but gender isn't.

Bottom line: When an interaction effect is significant, do not try to interpret the importance of main effects in isolation

Example

- Consider the following function:
- $$\ln Y = 6.89 + 0.5221 \ln X_1 + 0.6234 \ln X_2 - 0.2375 D_1 + 0.32445 D_2 + 0.6382 D_3$$
- $$(4.387)^{**} \quad (7.845)^{***} \quad (4.348)^{**} \quad (6.886)^{***} \quad (12.233)^{**} \quad (9.342)^{***}$$
$$R^2 = 0.766 \quad F = 15.234 \quad (0.0000982)$$

- Where Y = Labour hours of Company

X_1 = total assets of Company

X_2 = total number of offices in company

$D_1 = 1$ if management rating was “good”

$D_2 = 1$ if management rating was “fair”

$D_3 = 1$ if management rating was “satisfactory”

The figures in () are t values and ** - indicates significant at 5% level and *** - indicates significant at 10 % level of significance.

The management rating was taken on four point scale (Unfair, Good, Fair and Satisfactory).

- Interpret the results in context with X_1 and X_2 .
- How would you interpret the dummy coefficients.
- Discuss the concept of dummy variable trap.

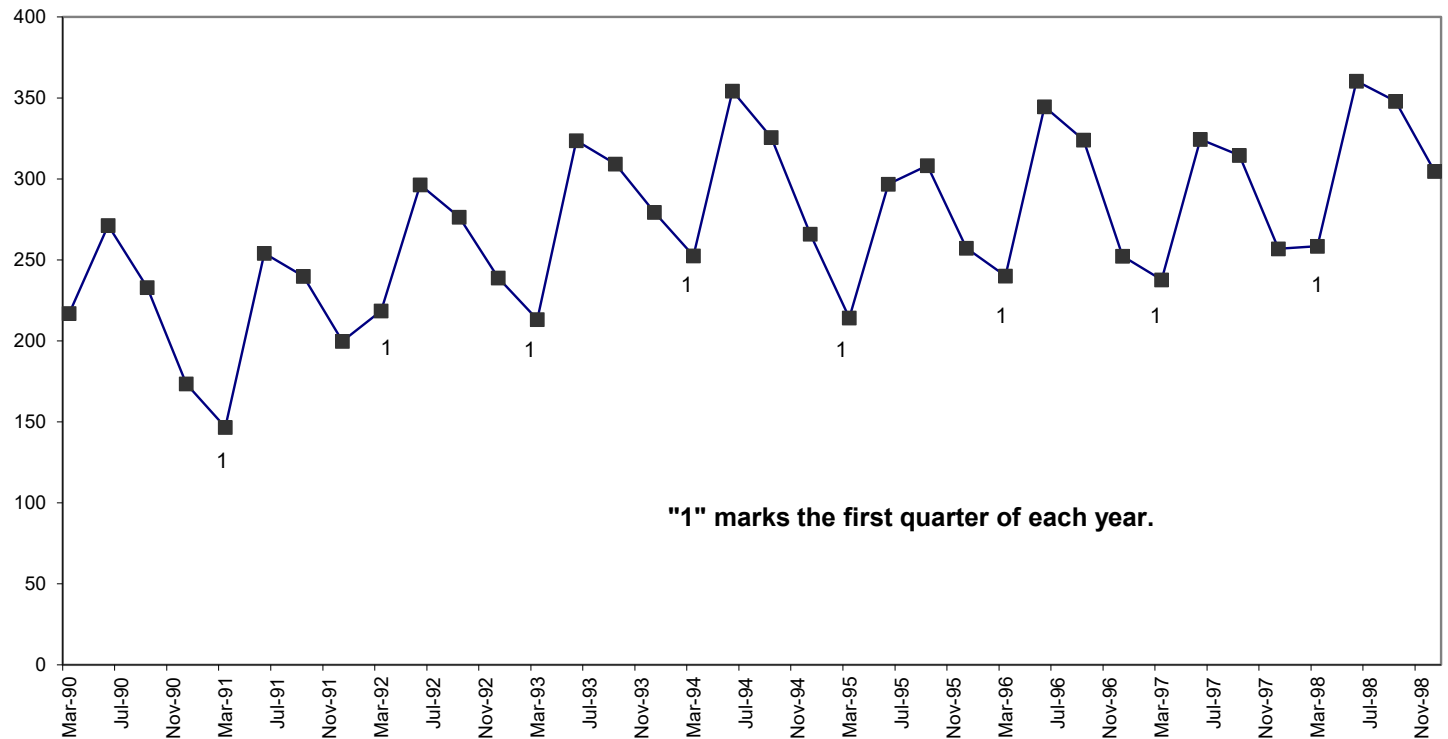
Use of Dummy Variables in Seasonal Analysis

- Seasonal Patterns are not easily accounted for by the typical causal variables that we use in regression analysis.
- An indicator variable can be used effectively to account for seasonality in our time series data.
- The number of seasonal indicator variables to use depends on the data.
- If we have p periods in our data series, **we can not use** more than $p-1$ seasonal indicator (dummy) variables.

Example: Private Housing Starts (PHS)

- Housing starts in the United States measured in thousands of units. These data are plotted for 1990 Q1 through 1999Q4. There are typically few housing starts during the first quarter of the year (January, February, March); there is usually a big increase in the second quarter of (April, May, June), followed by some decline in the third quarter (July, August, September), and further decline in the fourth quarter (October, November, December).

Private Housing Starts (PHS) in Thousands of Units



Example: Private Housing Starts (PHS)

- To Account for and measure this seasonality in a regression model, one can use three dummy variables: Q_2 for the second quarter, Q_3 for the third quarter, and Q_4 for the fourth quarter. These will be coded as follows:
 - $Q_2 = 1$ for all second quarters and zero otherwise.
 - $Q_3 = 1$ for all third quarters and zero otherwise
 - $Q_4 = 1$ for all fourth quarters and zero otherwise.

Example: Private Housing Starts (PHS)

- Data for private housing starts (PHS), the mortgage rate (MR), and these seasonal indicator variables are shown on the next slide.
- Examine the data carefully to verify your understanding of the coding for Q2, Q3, Q4.
- Since we have assigned dummy variables for the second, third, and fourth quarters, the first quarter is the base quarter for our regression model.
- Note that any quarter could be used as the base, with indicator variables to adjust for differences in other quarters.

Example: Private Housing Starts (PHS)

PERIOD	PHS	MR	Q2	Q3	Q4
31-Mar-00	217	10.1202	0	0	0
30-Jun-00	271.3	10.3372	1	0	0
30-Sep-00	233	10.1033	0	1	0
31-Dec-00	173.6	9.9547	0	0	1
31-Mar-01	146.7	9.5008	0	0	0
30-Jun-01	254.1	9.5265	1	0	0
30-Sep-01	239.8	9.2755	0	1	0
31-Dec-01	199.8	8.6882	0	0	1
31-Mar-02	218.5	8.7098	0	0	0
30-Jun-02	296.4	8.6782	1	0	0
30-Sep-02	276.4	8.0085	0	1	0
31-Dec-02	238.8	8.2052	0	0	1
31-Mar-03	213.2	7.7332	0	0	0
30-Jun-03	323.7	7.4515	1	0	0
30-Sep-03	309.3	7.0778	0	1	0
31-Dec-03	279.4	7.0537	0	0	1
31-Mar-04	252.6	7.2958	0	0	0
30-Jun-04	354.2	8.4370	1	0	0
30-Sep-04	325.7	8.5882	0	1	0
31-Dec-04	265.9	9.0977	0	0	1
31-Mar-05	214.2	8.8123	0	0	0
30-Jun-05	296.7	7.9470	1	0	0
30-Sep-05	308.2	7.7012	0	1	0
31-Dec-05	257.2	7.3508	0	0	1
31-Mar-06	240	7.2430	0	0	0
30-Jun-06	344.5	8.1050	1	0	0
30-Sep-06	324	8.1590	0	1	0
31-Dec-06	252.4	7.7102	0	0	1
31-Mar-07	237.8	7.7905	0	0	0
30-Jun-07	324.5	7.9255	1	0	0
30-Sep-07	314.6	7.4692	0	1	0
31-Dec-07	256.8	7.1980	0	0	1
31-Mar-08	258.4	7.0547	0	0	0
30-Jun-08	360.4	7.0938	1	0	0
30-Sep-08	348	6.8657	0	1	0
31-Dec-08	304.6	6.7633	0	0	1
31-Mar-09	294.1	6.8805	0	0	0
30-Jun-09	377.1	7.2037	1	0	0
30-Sep-09	355.6	7.7990	0	1	0
31-Dec-09	308.1	7.8338	0	0	1

Example: Private Housing Starts (PHS)

- The regression model for private housing starts (PHS) is:

$$PHS = \beta_0 + \beta_1(MR) + \beta_2(Q2) + \beta_3(Q3) + \beta_4(Q4)$$

- In this model we expect b_1 to have a negative sign, and we would expect b_2, b_3, b_4 all to have positive signs. Why?
- Regression results for this model are shown in the next slide.

Example: Private Housing Starts (PHS)

SUMMARY OUTPUT						
<i>Regression Statistics</i>						
Multiple R	0.885398221					
R Square	0.78393001					
Adjusted R Square	0.759236296					
Standard Error	26.4498851					
Observations	40					
ANOVA						
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	4	88837.93624	22209.48406	31.74613731	3.33637E-11	
Residual	35	24485.87476	699.5964217			
Total	39	113323.811				
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	473.0650749	35.54169837	13.31014264	2.93931E-15	400.9115031	545.2186467
MR	-30.04838192	4.257226391	-7.058206249	3.21421E-08	-38.69102153	-21.40574231
Q2	95.74106935	11.84748487	8.081130334	1.6292E-09	71.689367	119.7927717
Q3	73.92904763	11.82881519	6.249911462	3.62313E-07	49.91524679	97.94284847
Q4	20.54778131	11.84139803	1.73524961	0.091495355	-3.491564078	44.5871267

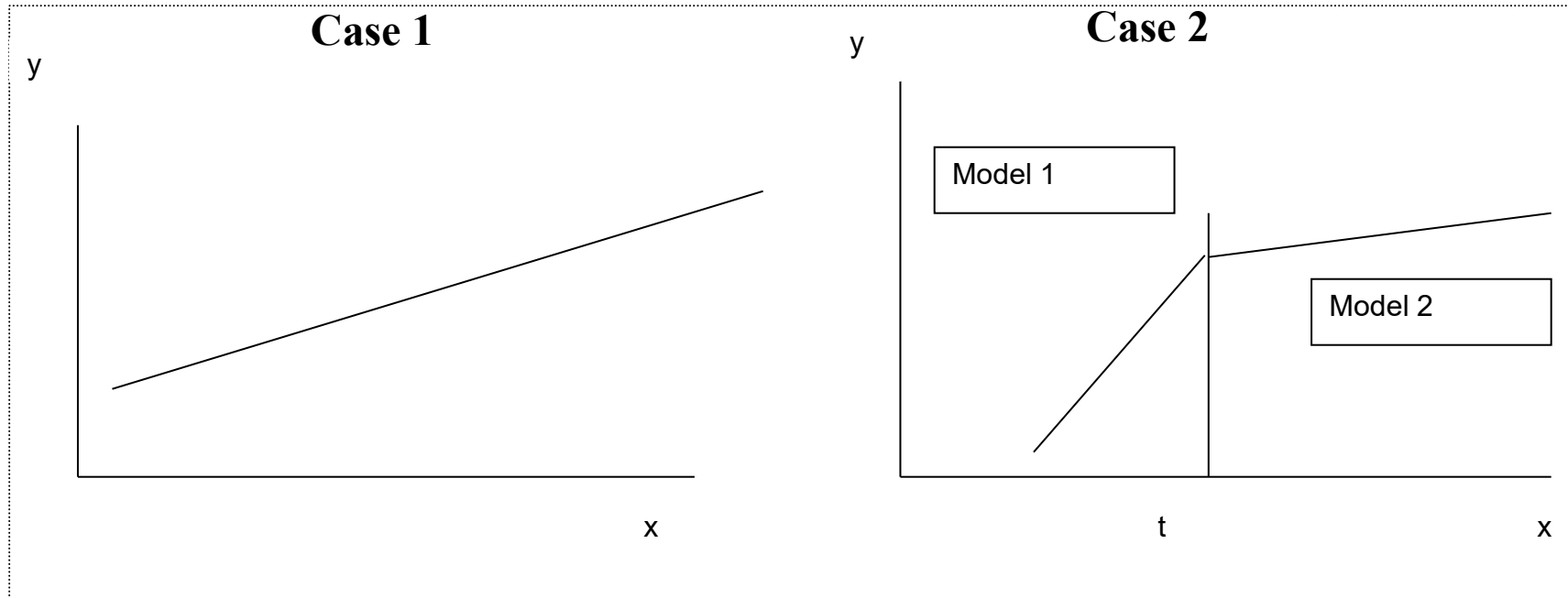
Example: Private Housing Starts (PHS)

- Use the prediction equation to make a forecast for each of the fourth quarter of 2009.
- Prediction equation:

$$PH\hat{S} = 473.06 - 30.05(MR) + 95.74(Q2) + 73.93(Q3) + 20.55(Q4)$$

Testing structural stability of regression models

- A series of data can often contain a **structural break**, due to a change in policy or sudden shock to the economy, e.g., 2008 Lehman Brother crisis and stock market crash.



In the first case we have just a single regression line to fit the data points (scatterplot), it can be expressed as:

$$y_t = \alpha_0 + \alpha_1 x_t + u_t$$

In the second case, where there is a structural break, we have two separate models, expressed as:

$$y_t = \beta_1 + \beta_2 x_t + u_{1t}$$

$$y_t = \delta_1 + \delta_2 x_t + u_{2t}$$

Testing structural stability of regression models

- This suggests that model 1 applies before the break at time t , then model 2 applies after the structural break.
- If the parameters in the above models are the same, i.e. $\beta_1 = \delta_1$, $\beta_2 = \delta_2$, then models 1 and 2 can be expressed as a single model as in case 1, where there is a single regression line.
- **The Chow test** basically tests whether the single regression line or the two separate regression lines fit the data best.
- The stages in running the Chow test are explained on next slide:

Chow Test for Testing Structural Stability – Steps

- **Step 1:** Firstly run the regression using all the data, before and after the structural break, collect RSS_c .
- **Step 2:** Run two separate regressions on the data before and after the structural break, collecting the RSS in both cases, giving RSS_1 and RSS_2 .
- **Step 3:** Using these three values, calculate the test statistic from the following formula:

$$F = \frac{RSS_c - (RSS_1 + RSS_2) / k}{RSS_1 + RSS_2 / n - 2k}$$

- **Step 4:** Find the critical values in the F-test tables, in this case it has $F(k, n-2k)$ degrees of freedom.
- **Step 5:** Conclude, the null hypothesis is that there is no structural break.

Piecewise Linear Regression

- A piecewise linear regression is made up of pieces of two linear regression lines joined at the knot or the threshold value
 - Could also have multiple piecewise linear regressions (with multiple knots)
 - Type of spline function
 - Regression represented as:

$$Y_i = B_0 + B_1 X_i + B_2 (X_i - X^*) D_i + u_i$$

where X^* represents the threshold value and

$$D = 1 \quad \text{if } X^* < X \quad \quad \text{and} \quad \quad D = 0 \quad \text{if } X^* > X.$$

Qualitative Response Regression Models

- The qualitative response models are considered as the binary response regression model.
- There are four approaches to developing a probability model for a binary response variable:
 - 1) The **linear probability model (LPM)**
 - 2) The **logit model**
 - 3) The **probit model**
 - 4) The **tobit model**

1) The Linear Probability Model (LPM)

- Consider the following regression model:

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

Where X = family income and

$Y = 1$ if the family owns a house and 0 if it does not own a house.

- The above model looks like a typical linear regression model but because the regressand is binary or dichotomous, it is called a linear probability model (LPM).

1) The Linear Probability Model (LPM)

- If P_i = probability that $Y_i = 1$ (the event occurs) and $(1-P_i)$ = probability that $Y_i = 0$ (the event does not occur), the variable Y_i has the following probability distribution:

Y_i	Probability
0	$1 - P_i$
1	P_i
Total	1

- That is, Y_i follows the Bernoulli probability distribution.

➤ **Non-normality of the disturbances u_i**

The assumption of normality for u_i is not tenable for the LPMs because, like Y_i , the disturbance u_i also take only two values, that is, they also follow the Bernoulli distribution.

1) The Linear Probability Model (LPM)

➤ Heteroscedastic Variance of the disturbances u_i

- Even if $E(u_i) = 0$ and $\text{cov}(u_i, u_j) = 0$ for $i \neq j$ (i.e., no serial correlation), it can no longer be maintained that in the LPM the disturbances are homoscedastic.
- To resolve this issue, we can use White's heteroscedasticity-corrected standard errors to deal with heteroscedasticity.

➤ Questionable value of R^2 as a Measure of Goodness of Fit

- The conventionally computed R^2 is likely to be much lower than 1 for such models. In most practical applications the R^2 ranges between 0.2 to 0.6. It can even take a higher value. Hence John Aldrich and Forrest Nelson contend that “use of the coefficient of determination as a summary statistic should be avoided in models with qualitative dependent variables.”

2) The Logit Model

- Consider the following representation of home ownership. The probability of home ownership in relation to income is given by

$$P_i = \frac{1}{1 + e^{-(\beta_1 + \beta_2 X_i)}}$$

- For ease of exposition, we write,

$$P_i = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

Where $z = \beta_1 + \beta_2 X_i$

$$1 - P_i = \frac{1}{1 + e^z}$$

$$\text{Hence, } \frac{P_i}{1 - P_i} = e^z \longrightarrow \text{Odds ratio}$$

2) The Logit Model

- Taking natural log of the expression, we get,

$$\begin{aligned} L_i &= \ln \left(\frac{P_i}{1 - P_i} \right) = Z_i \\ &= \beta_1 + \beta_2 X_i \end{aligned}$$

- L_i , the log of odds ratio, is called the logit, and hence the name logit model for the models mentioned above.

2) The Logit Model - Interpretation

- Let us consider the following hypothetical example:

$$\text{logit}(p) = \ln \left(\frac{P_i}{1 - P_i} \right) = 0.5 + 0.13 * \text{study_hours} + 0.97 * \text{female}$$

- In the model above, $b = 0.13$, $c = 0.97$, and $p = P\{Y=1\}$ is the probability of passing a math exam.
- Increasing the study hours by 1 unit (1 hour) will result in a 0.13 increase in $\text{logit}(p)$ or $\log(p/1-p)$. Now, if $\log(p/1-p)$ increases by 0.13, that means that $p/(1 - p)$ will increase by $\exp(0.13) = 1.14$. This is a 14% increase in the odds of passing the exam (*assuming that the variable female remains fixed*).
- Let's also interpret the impact of being a female on passing the exam. We know that $\exp(0.97) = 2.64$. That being said, the odds for passing the exam are 164% higher for women.

3) Probit Model

- If response variable is home ownership, the decision of the i^{th} family to own a house or not depends on an unobservable utility index I_i (also known as a latent variable), that is determined by one or more explanatory variables, say income X_i , in such a way that the larger the value of the index I_i , the greater the probability of a family owning a house.
- The index is expressed as,

$$I_i = \beta_1 + \beta_2 X_i$$

Where X_i is the income of the i^{th} family.

- Given the assumption of normality, the probability that I_i^* is less than or equal to I , can be computed from the standardized normal cumulative distribution function (CDF) as:

$$P_i = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{I_i} e^{-z^2/2} dz = P(Z_i \leq \beta_1 + \beta_2 X_i) = F(\beta_1 + \beta_2 X_i)$$

- Hence, if we want to find the effect of unit change in X on the probability that $P = 1$, we need to take the derivative of this function with respect to X .

$$\frac{dP_i}{dX_i} = f(\beta_1 + \beta_2 X_i) \beta_2$$

Calculate Z value corresponding to $(\beta_1 + \beta_2 X_i)$ from Z table and multiply it by β_2 to obtain the probability of owning house given X .

4) Tobit Model

- Extension of Probit model developed by James Tobin, the Noble Laureate Economist.
- Introduced a dilemma: If a consumer does not purchase a house, obviously we have no data on housing expenditure for such consumers; we have such data only on consumers who actually purchase a house.
- Thus, the consumers are divided into two groups, one consisting of say n_1 consumers about whom we have information on the regressor (Income) as well as regressand (Ownership), and another consisting of n_2 consumers about whom we have information only on the regressor but not the regressand.
- The second sample is known as censored sample, and hence the tobit model is also known as a censored regression model or limited dependent variable regression model, expressed as below:

$$\begin{aligned} Y_i &= \beta_1 + \beta_2 X_i + u_i && \text{if RHS} > 0 \\ &= 0 && \text{otherwise.} \end{aligned}$$

The results are interpreted similar to probit model.